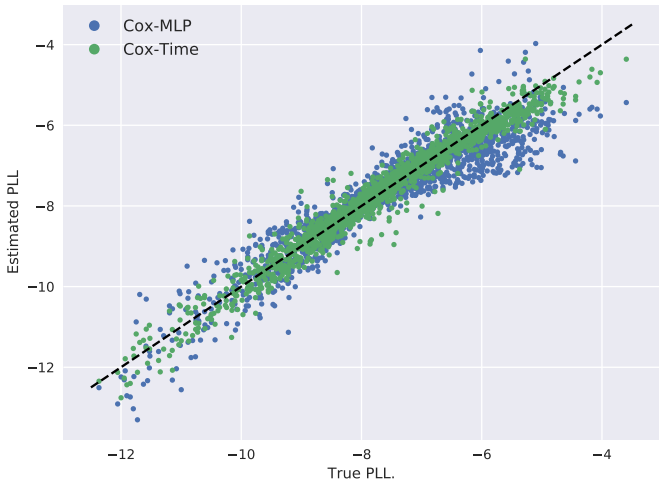


Figure C.2: Partial log-likelihood estimates of Cox-MLP and Cox-Time plotted against the true partial log-likelihoods, for 2,000 samples of the test set.

were $g_{ph}(\mathbf{x})$ is the function in (C.1). The simulations are otherwise unchanged from the previous experiments. We require the term $b(\mathbf{x})$ to be non-negative to ensure that the hazards increase with time. Furthermore, we have added $\text{sign}(x_3)$ to $a(x)$ as this is essentially equivalent to having two different baselines, $\tilde{h}_0 = h_0 \exp[\text{sign}(x_3)] \in \{0.0074, 0.054\}$. Both of these choices are motivated by our attempt to produce reasonable looking survival curves.

We sample 10,000 training samples, 10,000 test samples, and 2,000 samples for validation, and fit a Cox-MLP model and a Cox-Time model. Both Cox-MLP and Cox-Time parameterize g with a 4 hidden layer MLP with ReLU activations, 128 nodes in each layer, and dropout between layers with rate 0.1.

In Figure C.2, we have plotted the estimated partial log-likelihoods of Cox-MLP and Cox-Time against the true partial log-likelihoods for 2,000 samples of the test set. Though the difference between the methods is not very large, Cox-Time appears to capture the true values better than Cox-MLP.

To further illustrate some of the differences between Cox-MLP and Cox-Time, we have in Figure C.3 plotted nine survival curves $\hat{S}(t \,|\, \mathbf{x})$ from the test set. The figure shows both the true curves and the curves estimated by the two methods. It is seen that Cox-Time is able to estimate the true survival curves better than those produced by Cox-MLP.

C.3 Simulation Details

In the following, we explain in detail how we generated our simulated data sets in Section 5 and Appendices C.1 and C.2. We want to generate survival times T^* from relative risk models of the form

$$h(t \,|\, \mathbf{x}) = h_0(t) \exp[g(t, \mathbf{x})]. \tag{C.2}$$

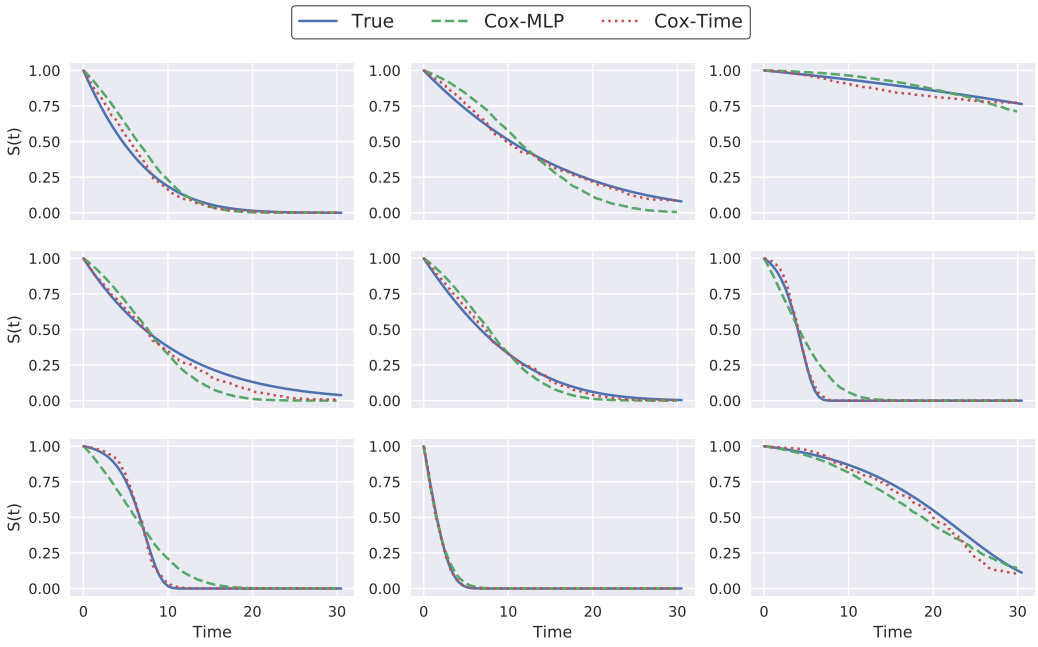


Figure C.3: Examples of survival curves $\hat{S}(t \,|\, \mathbf{x})$ from the test set in the non-proportional simulation study.

If the function g does not depend on t, we have a proportional hazards model, which is just a special case of the relative risk models. Let $H(t \,|\, \mathbf{x})$ denote the continuous (increasing) cumulative hazard $H(t \,|\, \mathbf{x}) = \int_0^t h(s \,|\, \mathbf{x})\, ds$, and let V be exponentially distributed with parameter 1. Then we can obtain survival times through the inverse cumulative hazard $T^* = H^{-1}(V \,|\, \mathbf{x})$. This can be shown to hold through the hazard's relationship to the survival function in (1),

$$S(t \,|\, \mathbf{x}) = \mathrm{P}(T^* > t \,|\, \mathbf{x}) = \mathrm{P}(H^{-1}(V \,|\, \mathbf{x}) > t) = \mathrm{P}(V > H(t \,|\, \mathbf{x})) = \exp[-H(t \,|\, \mathbf{x})],$$

as V is exponentially distributed with $\mathrm{P}(V > v) = \exp(-v)$. Hence, we can obtain survival times by transforming generated samples from an exponential distribution.

To obtain an analytical expression for the inverse cumulative hazard $H^{-1}(v \,|\, \mathbf{x})$, we restrict the models in (C.2) to have constant baseline h_0, in addition to a simple time dependence in $g(t, \mathbf{x})$. For the linear predictor in Section 5 and the non-linear predictor in Appendix C.1 we simulate with proportional hazards, meaning $g(t, \mathbf{x}) = g(\mathbf{x})$. Hence, we get cumulative hazards and inverse cumulative hazards of the form

$$\begin{aligned} H(t \,|\, \mathbf{x}) &= t\, h_0 \exp[g(\mathbf{x})], \\ H^{-1}(v \,|\, \mathbf{x}) &= \frac{v}{h_0 \exp[g(\mathbf{x})]}. \end{aligned}$$

In the non-proportional simulations in Appendix C.2, we add a time dependent term $g(t, \mathbf{x}) = a(\mathbf{x}) + b(\mathbf{x})\, t$, which gives us

$$H(t \,|\, \mathbf{x}) = \frac{h_0 \exp[a(\mathbf{x})]\,(\exp[b(\mathbf{x})\, t] - 1)}{b(\mathbf{x})},$$
$$H^{-1}(v \,|\, \mathbf{x}) = \frac{1}{b(\mathbf{x})} \log\left(1 + \frac{v\, b(\mathbf{x})}{h_0 \, \exp[a(\mathbf{x})]}\right).$$

Appendix D. DeepHit

DeepHit by Lee et al. (2018) is a discrete survival model for competing risks. However, as we only consider one type of event, we will express the method in terms of a single cause. DeepHit considers time to be discrete, so to fit it to the continuous-time data sets in Section 6, we discretize the event times with an equidistant grid between the smallest and largest duration in the training set. The number of discrete time-points is considered a hyperparameter, given by "Num. durations" in Table A.1.

Now, assume time is discrete with $0 = \tau_0 < \tau_1 < \tau_2 < \cdots < \tau_m$. Let $\mathbf{y}(\mathbf{x}) = [y_0(\mathbf{x}), y_1(\mathbf{x}), \ldots, y_m(\mathbf{x})]^T$ denote the output of a neural net with covariates $\mathbf{x}$ and a softmax output activation. Then, $\mathbf{y}(\mathbf{x})$ can be interpreted as the estimated probability mass function of the duration times $y_j(\mathbf{x}_i) = \hat{\mathrm{P}}(T_i^* = \tau_j \,|\, \mathbf{x}_i)$. The estimated survival function is then given by

$$\hat{S}(\tau_j \,|\, \mathbf{x}) = 1 - \sum_{k=1}^{j} y_k(\mathbf{x}),$$

and the discrete negative log-likelihood corresponding to the continuous version in (2), is

$$\text{loss}_L = -\sum_{i=1}^{N} \Big[D_i \log(y_{e_i}(\mathbf{x}_i)) + (1 - D_i) \log(\hat{S}[T_i \,|\, x_i]) \Big].$$

Here, e_i denotes the index of the event time for observation i, i.e., $T_i = \tau_{e_i}$. Furthermore, DeepHit adds a second loss that attempts to improve its ranking capabilities,

$$\text{loss}_{\text{rank}} = \sum_{i,j} D_i \, \mathbb{1}\{T_i < T_j\} \exp\left(\frac{\hat{S}(T_i \,|\, \mathbf{x}_i) - \hat{S}(T_i \,|\, \mathbf{x}_j)}{\sigma}\right). \tag{D.1}$$

The loss of DeepHit is a combination of these two losses, where the ranking loss is scaled by a constant. We deviate slightly from the original implementation here and instead use a convex combination of the two,

$$\text{loss} = \alpha \, \text{loss}_L + (1 - \alpha) \, \text{loss}_{\text{rank}}, \tag{D.2}$$

where α and σ from (D.1) are considered hyperparameters.

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